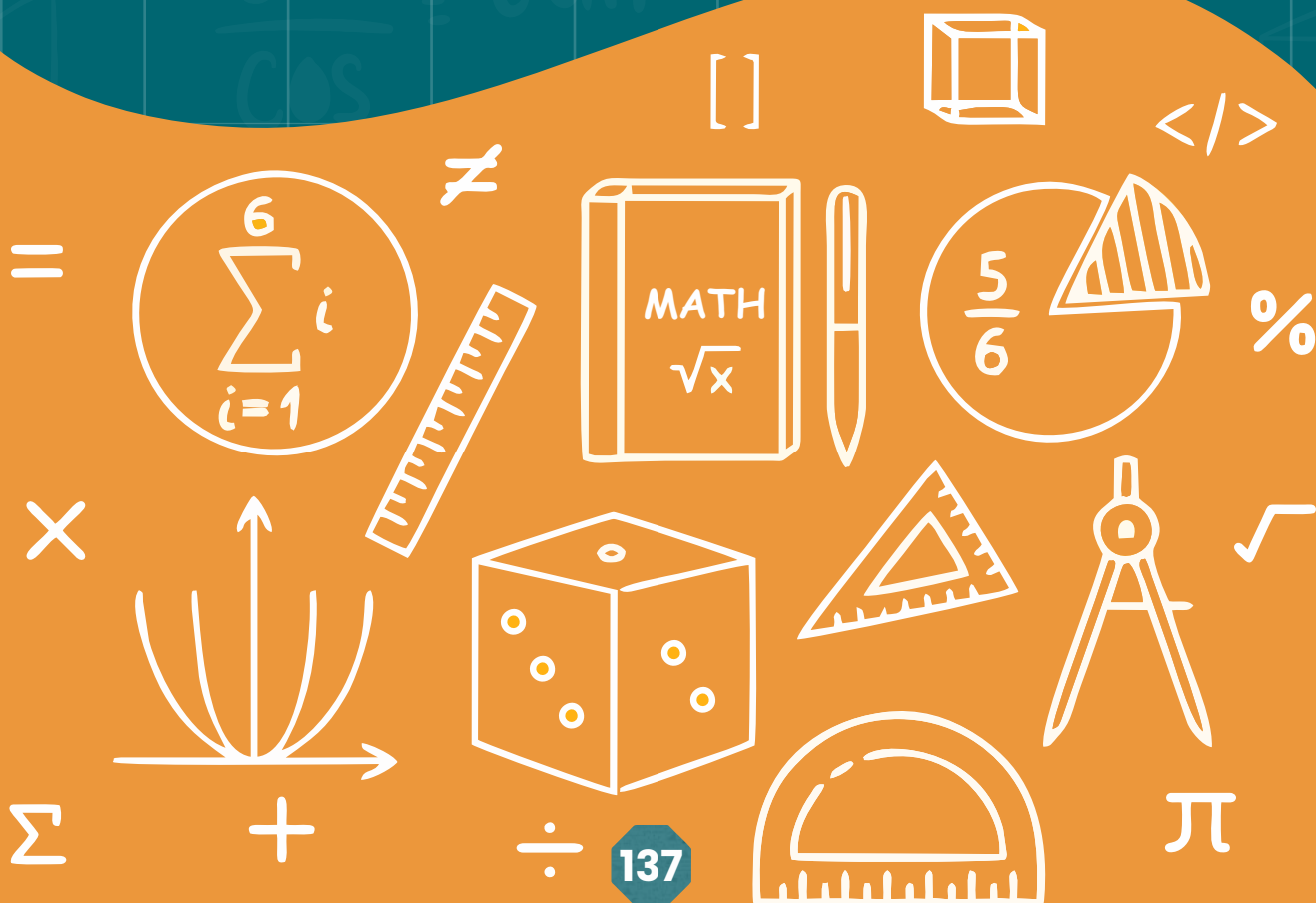


UNIT 12

CIRCUMFERENCE, AREA AND VOLUME



CIRCUMFERENCE AND AREA OF A CIRCLE

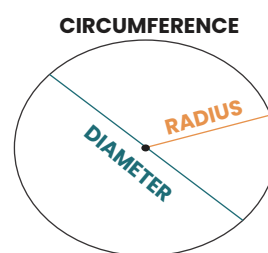
A circle is a shape where all points are equally distant from its centre. The circumference of a circle is the distance around its boundary, calculated using the formula $C = 2\pi r$, where r is the radius. The area of a circle is the space it covers, found using the formula $A = \pi r^2$. These formulas help us understand the size and boundary of a circle.

Circumference of a Circle

The circumference is the distance around the circle (its boundary). Think of it as the "perimeter" of the circle.

Formula for Circumference:

$$C = 2\pi r$$

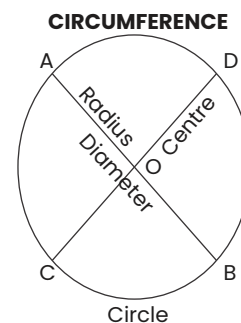


$$C = \pi d \quad \text{or} \quad C = 2\pi r$$

Where:

- C is the circumference.
- r is the radius (distance from the centre of the circle to any point on its edge).
- π is a constant whose value is approximately equal to $22/7$ or 3.142 .

Circle is a familiar shape. O is the centre of the circle. OA is radius. AB is diameter. Circle is a closed figure and every point on it is at the same distance from a given point O , the centre.



The measure of the length of the boundary of a circle is called its **circumference**.

Whereas $\overline{AOD}$, $\overline{BOD}$, $\overline{AOC}$, and $\overline{BOC}$ are sectors of the circle. $\overline{OA}$, $\overline{OB}$, $\overline{OC}$, and $\overline{OD}$ all represent the radius of the given circle.

AB and CD are diameters. Since a diameter is a line segment, it can easily be measured.

The table below shows the measure of diameters and approximate size of circumferences of four circles. Find the value of the ratio $\left(\frac{\text{Circumference}}{\text{Diameter}}\right)$ for each circle. What did you notice?

Circle	Diameter	Circumference (Approx. value)	$\frac{\text{Circumference}}{\text{Diameter}}$ (Approx. value)
(i)	3.5 cm	11 cm	
(ii)	28 cm	88 cm	
(iii)	4.2 cm	13.2 cm	
(iv)	21 cm	66 cm	

Example: The length of the diameter of a circle is 14 cm. Find the circumference of the circle?

Solution: The length of the diameter = 14 cm and $\pi = \frac{22}{7}$
 Thus, the circumference of the circle = $\pi \times d$
 $= \frac{22 \times 14}{7} = 44 \text{ cm}$

Example: Find the diameter and radius of the circle, when its circumference is 99 cm.

Solution: We know that:

$$\text{Circumference} = \pi \times d$$

$$\text{or } \frac{C}{\pi} = d$$

So,

$$\begin{aligned} \text{Diameter} = d &= C / \pi \\ &= 99 / 22/7 \\ &= 99 \times 7 / 22 \\ &= 31.5 \text{ cm} \end{aligned}$$

and

$$\begin{aligned} \text{radius} = r &= d / 2 \\ &= 31.5 / 2 \\ &= 15.75 \text{ cm} \end{aligned}$$

AREA OF CIRCLE

Area of a circle is the region occupied by the circle in a two-dimensional plane. It can be determined easily using a formula, $A = \pi r^2$, (Pi r-squared), where r is the radius of the circle. The unit of area is the square unit, such as m^2 , cm^2 , etc.

Example 1: The radius of a circle is 35 cm. Find its area.

Solution: Area of the circular region = $\pi \times r^2$

Here radius $r = 35$ cm, and $\pi = 22/7$

$$\text{Area} = \pi \times r^2 = \frac{22 \times 35 \times 35}{7} = 3,850 \text{ sq. cm}$$

Example 2: The area of a circular region is 1,386 sq. cm. Find its radius.

Solution: Given area A of the circular region is 1,386 sq. cm.

We know that the formula for finding the area of a circular region is: $A = \pi \times r^2$.

$$\text{Therefore, } 1386 = \pi \times r^2$$

$$\text{or } 22/7 \times r^2 = 1386$$

$$\text{or } r^2 = (1386 \times 7) / 22$$

$$\text{So, } r^2 = 63 \times 7 = 441$$

$$\text{Thus, radius } r = 21 \text{ cm}$$

EXERCISE 12.1

A. Find the circumference of the circle when its diameter is:

(1) $d = 28$ cm

(2) $d = 35$ cm

(3) $d = 42$ mm

(4) $d = 56$ mm

B. Find the circumference of the circle when its radius is:

(1) 10.5 cm

(2) 28 cm

(3) 38.5 cm

(4) 49 cm

C. Find the radius of the circle when its circumference is:

(1) 22 cm

(2) 66 cm

(3) 88 cm

(4) 110 cm

D. Find the diameter of the circle when its circumference is:

(1) 44 cm

(2) 154 cm

(3) 242 mm

(4) 264 mm

A. Find the area of the circular region when its radius is:

(1) 14 cm

(2) 10.5 cm

(3) 21 cm

(4) 17.5 mm

B. Find the radius of the circular region when its area is:

(1) 154 sq. mm

(2) 2,464 sq. cm

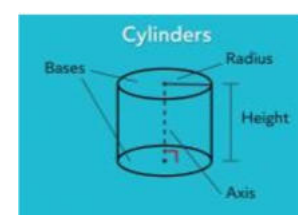
(3) 7,546 sq. mm

SURFACE AREA AND VOLUME OF A CYLINDER

To find the surface area and volume of a cylinder, you can use the following formulas:

Surface area: $2\pi rh + 2\pi r^2$

Volume: $\pi r^2 h$



In these formulas, r is the radius of the cylinder and h is its height. You can think of a cylinder's surface as a rectangle with a length equal to the circumference of the base and a width equal to the height of the cylinder. The total surface area is the sum of the areas of the two circles and the rectangle.

SURFACE AREA OF A CYLINDER

The surface area of a cylinder refers to the total area that the cylinder's curved surface and circular bases cover. The total surface area of the cylinder includes the area of the 2 bases of the cylinder (each of which is in the shape of a circle) and the area of the curved surface. Surface area is expressed in square units like square centimetres, square inches, square feet, and so on.

Total Surface Area of Cylinder = $2\pi r^2 + 2\pi rh$ (or) $2\pi r(r + h)$

Example:

Find the total surface area (TSA) of a cylinder of radius 5 cm and height 8 cm.

Solution: The total surface area (TSA) of a cylinder can be calculated using the formula,

$$TSA = 2\pi r(r + h).$$

By substituting the values of $r = 5$, $h = 8$, we get:

$$\begin{aligned} TSA &= 2\pi r(r + h) \\ &= 2 \times 3.14 \times 5(5 + 8) \\ &= 408.41 \text{ cm}^2 \end{aligned}$$

VOLUME OF A CYLINDER FORMULA

The volume of a cylinder is the space it occupies, calculated by finding the area of its circular base and multiplying it by its height. For a cylinder with base radius r and height h , the formula is: Volume = $\pi r^2 h$, where r^2 is the square of the radius, and h is the height. The result is measured in cubic units.

Examples

Calculate the volume of a given cylinder having height 20 cm and base radius of 14 cm. (Take $\pi = 22/7$)

Solution:

Given:

Height = 20 cm
radius = 14 cm

we know that; $V = \pi r^2 h$ cubic units

$$V = \left(\frac{22}{7}\right) \times 14 \times 14 \times 20$$

$$V = 12320 \text{ cm}^3$$

Therefore, the volume of a cylinder = 12320 cm^3

Solve real life problems involving Circumference and Area of Circle

Example

The diameter of a round table is 4.2 m. How many metres of lace are required for its border? Also find its cost at the rate Rs 40 per metre.

Solution:

First we have to find the circumference of the round table.

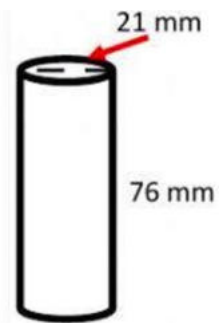
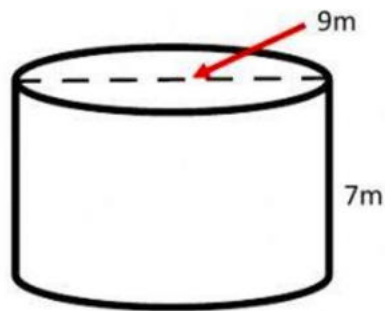
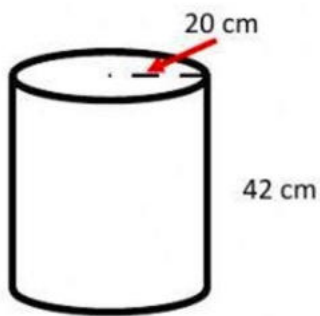
$$\text{Circumference} = \pi \times d = \frac{22}{7} \times 4.2 \text{ m} = \frac{22}{7} \times 4.2 = 13.2 \text{ m}$$

Thus the length of the lace required for rounding the circular cloth is 13.2 metres.

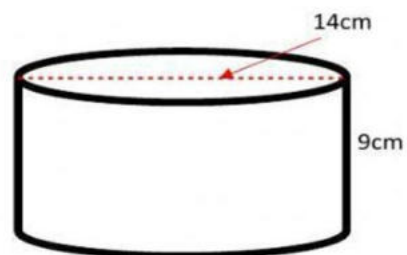
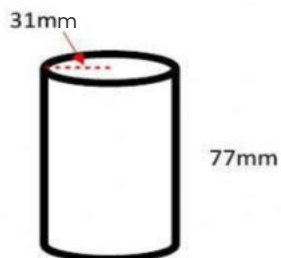
$$\begin{aligned}\text{The cost of the length of lace} &= \text{Rate} \times \text{length} \\ &= \text{Rs } 40 \times 13.2 \\ &= \text{Rs } 528\end{aligned}$$

EXERCISE 12.2

1. Find the Surface area for each cylinder. Use 3.14 for π and round all answers to the nearest tenth.



2. Find the Volume of each cylinder. Use 3.14 for π .



Volume: _____ mm^3 Volume: _____ cm^3

WORD PROBLEMS

- 1.** The diameter of a circular garden is 70 m. The garden has 5 rounds of wire fencing. If the wire costs Rs 10 per metre, find the total cost.
- 2.** Find the radius of a circular ring prepared from a wire used for a square with side 22 cm.
- 3.** The radius of a bicycle wheel is 42 cm. Each wheel completed 1000 rotations when Ahmed traveled from home to school. Find the distance.
- 4.** The total cost of the wire required for four rounds of a circular fencing is Rs 3,960/-. If the wire costs Rs 5 per metre, find the total length of wire used.